

# Two resolutions over 46 seconds

What the extended  $256^3$  run settles, and the one thing it unsettles

Report II — companion to `report_rot192_rot256`

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5 August 2026

## Abstract

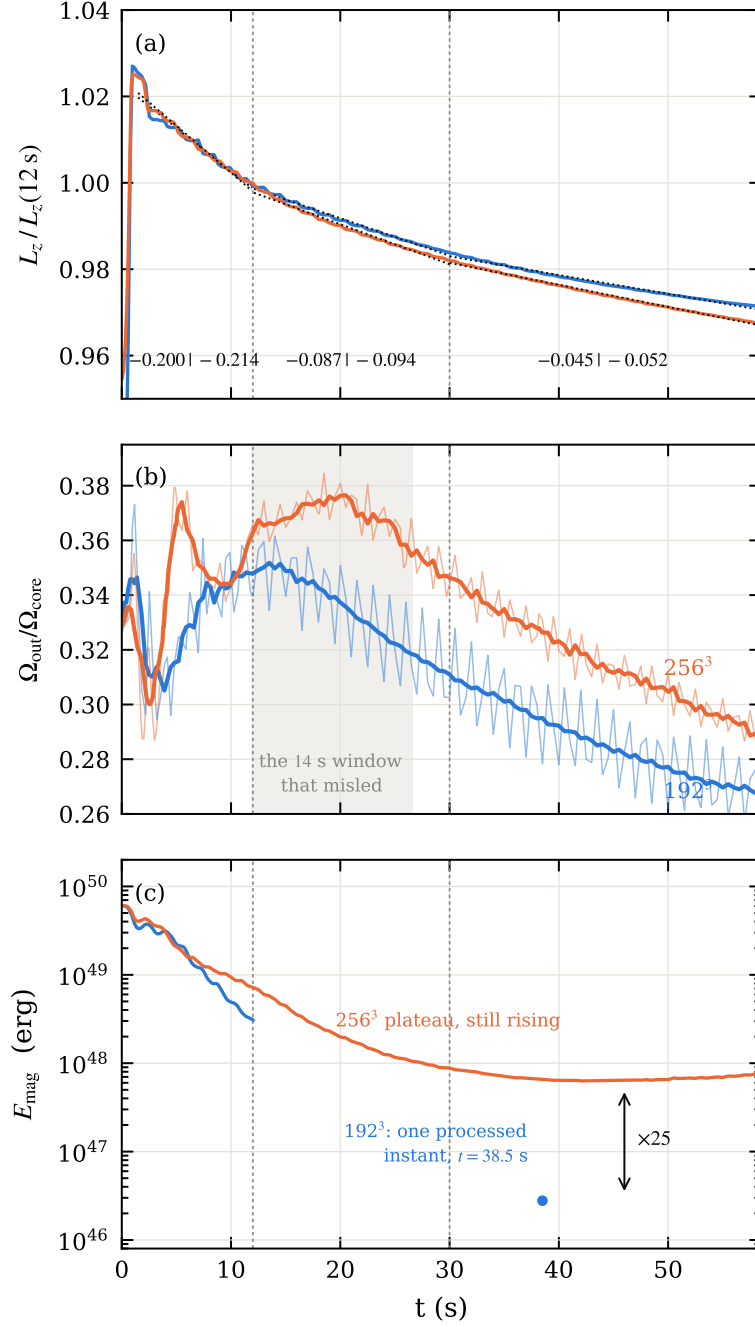
The first report compared the  $192^3$  and  $256^3$  runs at  $t = 12$  s. The  $256^3$  run has since been carried to  $t = 58.2$  s, so the comparison now spans 46 seconds rather than three. Three results follow. **The braking of  $L_z$  converges:** three successive regimes, each about half the rate of the one before, agreeing between the grids to 7, 7 and 15 percent — so the identification of the angular-momentum transport as magnetic, previously resting on one resolution, now rests on two. **The steepening of the differential rotation converges,** 22.6% against 20.3% over the full baseline. **The residual magnetic field does not:** it settles a factor of 25 higher on the finer grid and is still rising at  $t = 58$  s. A methodological result comes with the second: over a 14 s window the same two runs appear to disagree about the steepening by a factor of eight, because the  $256^3$  onset is some 6 s late. Two earlier drafts drew two different wrong conclusions from that window.

## 1 What this report adds

Report I established the configuration, the instability and the survival of the star, and compared the two grids over the three seconds where both existed. Its convergence statements were therefore all short-baseline statements. Since then the  $256^3$  run has been continued through twelve chained windows to  $t = 58.2$  s and all 164 plotfiles have been processed with the same diagnostics, so every quantity in Report I can now be asked of both grids over almost the whole evolution.

Nothing about the configuration, the code or the diagnostics has changed. The star is the same  $2.005 M_\odot$  differentially rotating equilibrium with a toroidal-dominated field; the tool is the same `tools/fbtbp.cpp` rebuilding everything from  $B_x, B_y, B_z$ .

Throughout, comparisons are made between means over two pulsation periods at each end of the baseline, never between single snapshots. The star breathes by  $\pm 14\%$  every 1.498 s, and a snapshot catches an arbitrary phase of it.



**Figure 1:** The three results. (a)  $L_z$  normalised at  $t = 12 \text{ s}$ , with the three braking regimes fitted; the numbers along the bottom are  $192^3 | 256^3$  in percent per second, and the dashed verticals are the regime boundaries. (b) The differential rotation. The finer grid begins to steepen about 6 s late, then steepens faster; the shaded band is the 14 s window over which that lag reads as a disagreement. (c) Magnetic energy, the one panel that does not converge. The  $192^3$  field diagnostics stop at  $t = 12 \text{ s}$ ; the isolated point at  $t = 38.5 \text{ s}$  is the single late instant that was processed, and is drawn as a marker because that is all it is.

## 2 The braking of $L_z$ converges

This is the strongest result of the comparison, and it is the one that matters most for the physics, because it is what identifies the transport.

There is no explicit viscosity and no explicit resistivity in these runs, so nothing else in the scheme brakes anything. The star loses angular momentum twice as fast while the ordered toroidal field exists as it does afterwards, and half as fast again once even the decaying field has gone.

| $d \ln L_z / dt$                |                      | 192 <sup>3</sup> | 256 <sup>3</sup> | agreement |
|---------------------------------|----------------------|------------------|------------------|-----------|
| $t = 1.5\text{--}11.9\text{ s}$ | toroidal field alive | $-0.1996\%/s$    | $-0.2144\%/s$    | 7%        |
| $t = 12\text{--}30\text{ s}$    | field decaying       | $-0.0873\%/s$    | $-0.0935\%/s$    | 7%        |
| $t = 30\text{--}58\text{ s}$    | residual             | $-0.0446\%/s$    | $-0.0523\%/s$    | 15%       |

**Table 1:** Three regimes, each braking at about half the rate of the one before, and the two grids agree on all three.

That factor of two, repeated, is the signature of magnetic transport, and it is now measured at two resolutions instead of one.

The third regime is new here — Report I described two. With the baseline extended it is clear that the braking does not simply switch from “fast” to “slow” when the field dies but continues to weaken as the field weakens, which is what a magnetic mechanism should do and what a numerical one need not.

**What this does not establish.** The rates agree between the grids; that does not make them physical. Both runs brake through whatever transport the scheme supplies, and if a component of that is numerical it may well converge at the same rate as the physical part. The argument for the magnetic identification is the *correlation with the field*, not the convergence; convergence only removes the worry that the correlation was a coarse-grid accident.

### 3 The steepening converges — over a long enough baseline

The differential rotation is the property the configuration depends on: a  $2 M_\odot$  white dwarf exists here only because  $\Omega$  falls steeply outwards. Report I found that the profile not only survives but steepens, and flagged the amplitude as unconverged.

| 12–15 s $\rightarrow$ 55–58.3 s            | 192 <sup>3</sup> | 256 <sup>3</sup> |                        |
|--------------------------------------------|------------------|------------------|------------------------|
| $L_z$                                      | $-2.6\%$         | $-2.9\%$         | converged              |
| $\Omega_{\text{out}}$                      | $-15.3\%$        | $-14.9\%$        | converged (3%)         |
| $\Omega_{\text{out}}/\Omega_{\text{core}}$ | $-22.6\%$        | $-20.3\%$        | <b>converged</b> (10%) |
| $\Omega_{\text{core}}$                     | $+9.4\%$         | $+6.7\%$         | same sign              |
| $L_z^{\text{outer}}/L_z^{\text{inner}}$    | $-10.6\%$        | $-6.2\%$         | same sign              |

**Table 2:** The steepening over 46 s. The amplitude converges to 10% and the outer-shell spin-down to 3%.

#### 3.1 A convergence test shorter than the lag measures the lag

The amplitude looked irreducibly resolution-dependent over shorter windows, and the reason is worth recording because it is a trap that will recur in this problem.

Neither number is wrong. The short baseline simply does not contain the onset, and a delayed effect measured over a window shorter than its delay is indistinguishable from an absent effect — and, if the lagging curve happens to drift the other way first, from an effect of the opposite sign.

This cost two wrong conclusions in successive drafts of Report I: first that the steepening was a coarse-grid artefact, then that its direction was real but its amplitude was not. Both were stated with the numbers that supported them, and both were wrong for the same reason.

The general form is uncomfortable, because the lag is not known in advance and is not obviously bounded. Here it is 6 s against a 1.5 s pulsation and a 2.4 s Alfvén time — a few dynamical times,

| change in $\Omega_{\text{out}}/\Omega_{\text{core}}$ | $192^3$   | $256^3$           |
|------------------------------------------------------|-----------|-------------------|
| $t = 12 \rightarrow 18$ s                            | $-3.2\%$  | $+\mathbf{2.1\%}$ |
| $t = 18 \rightarrow 24$ s                            | $-5.1\%$  | $-3.0\%$          |
| $t = 24 \rightarrow 30$ s                            | $-4.5\%$  | $-\mathbf{5.6\%}$ |
| $t = 30 \rightarrow 38$ s                            | $-3.8\%$  | $-\mathbf{5.2\%}$ |
| total over a 14 s baseline                           | $-8.2\%$  | $-1.0\%$          |
| total over a 46 s baseline                           | $-22.6\%$ | $-20.3\%$         |

**Table 3:** The finer grid is still flat — rising, even — while the coarse grid has already begun, and then steepens faster. The last two rows are the same comparison over two baselines: a factor of eight apart at 14s, ten percent apart at 46s.

which is reassuring but is a measurement, not a principle. Any future convergence test in this problem needs a baseline long enough that a plausible lag is a small fraction of it.

## 4 The residual field does not converge

The one place where extending the baseline made the disagreement worse.

| late-time field                | 192 <sup>3</sup>          | 256 <sup>3</sup>        | ratio         |
|--------------------------------|---------------------------|-------------------------|---------------|
| $E_{\text{mag}}$ plateau (erg) | $\simeq 3 \times 10^{46}$ | $7.0 \times 10^{47}$    | <b>25</b>     |
| $\max B $ (G)                  | $\simeq 1 \times 10^{12}$ | $6 \times 10^{12}$      | 6             |
| trend for $t > 45$ s           | —                         | $+0.012 \text{ s}^{-1}$ | <i>rising</i> |

**Table 4:** The 192<sup>3</sup> values are from the single late instant processed with the field diagnostics,  $t = 38.5$  s; the 256<sup>3</sup> values are means over  $t = 54$ – $58.3$  s. The trend is fitted to  $\ln E_{\text{mag}}$  over  $t > 45$  s, with  $E_{\text{tor}}$  and  $E_{\text{pol}}$  both rising.

This is consistent with what Report I already found: with no explicit resistivity, the decay of  $E_{\text{tor}}$  is the numerical resistivity, and it scales as  $dx^{2.4}$ . Less numerical dissipation leaves more field. But the consequences are sharper than the factor-of-two disagreement seen at  $t = 12$  s:

- The “disordered residual of order  $10^{12}$  G” reported at 192<sup>3</sup> describes a floor set by the mesh, not a state of the star.
- The factor 2170 by which  $E_{\text{mag}}$  falls at 192<sup>3</sup> is an **upper bound** on the destruction, and not a tight one.
- The 256<sup>3</sup> run has not reached a final state at all. A run that is gaining magnetic energy when it stops has not finished.

**Two readings, and two resolutions cannot separate them.** The mechanism for genuine regeneration is available and it is not exotic: the differential rotation survives and is still steepening, so there is free shear energy, and winding residual poloidal field into toroidal is the most ordinary thing a shear flow can do to a magnetic field. On that reading the finer grid is showing the beginning of a shear dynamo that the coarse grid dissipates before it can start.

The other reading is that both runs are on the same decaying track and the finer one is simply further from the mesh floor, with the apparent rise being the last decade of a decay it cannot resolve, contaminated by the pulsation.

The two look identical in the data we have. Distinguishing them needs either a third resolution — 384<sup>3</sup>, about five times the cost of 256<sup>3</sup> — or, better, explicit resistivity, which would give the problem a physical dissipation scale instead of letting the mesh set it. CASTRO does not currently implement it.

## 5 What changes in Report I

1. **Strengthened.** The magnetic identification of the transport. Two grids, three regimes, agreement of 7–15%.
2. **Strengthened.** The steepening of the differential rotation, now converged in amplitude as well as direction.
3. **Weakened.** Everything about the end state of the field. The residual is a mesh floor and the destruction factor is an upper bound. Report I’s Table 4 should be read with that in mind: its mass, radii and rotation rows are measurements, its field rows are bounds.
4. **New caveat.** Convergence tests in this problem need baselines longer than the inter-grid lag, and the lag is only known afterwards.

## Status and next steps

192<sup>3</sup>: complete,  $t = 60$  s, field diagnostics processed to  $t = 12$  s plus the single instant at 38.5 s.  
256<sup>3</sup>:  $t = 58.2$  s, all 164 plotfiles processed. A final window to 60 s is queued; nothing here depends on it.

The cheapest thing that would sharpen this report is not more physical time but the missing 192<sup>3</sup> field data: processing its plotfiles between  $t = 12$  and 60 s would turn the single point in Fig. 1c into a curve and make the residual comparison a proper one rather than a snapshot against a mean. It costs no new simulation.

Beyond that, the run that would test the interpretation directly is a genuinely mixed field — comparable poloidal and toroidal energy, rather than the  $10^{-7}$  ratio used here. A toroidal-dominated field is the textbook unstable case and behaved like one; the twisted-torus configuration is the one expected to survive, and it is the geometry the equilibrium literature on super-Chandrasekhar white dwarfs does not construct.

Inputs, submission scripts, diagnostic tools, data and figure code are under version control in the WD\_MAG repository. The data behind every number here is `investigations/bt_bp_256_long.csv` (164 rows,  $t = 0$  to 58.2 s) and `investigations/rotation_192.csv`.